Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes*

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March 3, 2026

Abstract

The structure of my intro should be the following paragraphs. (1) Hook, introduce ownership changes and nursing homes. Introduce research question (2) Why do nursing homes matter and why are they good as a way to study this. (3) Why is staffing the main input to nursing home production. (4) Why is staffing so important to patients and quality of care. (5) Two types of ownership changes, talk about type 1. (6) Type 2, this is the type we study. (7 & 8) Conceptual framework and the cost quality trade off. (9) Measurement, Restate Thesis Question and maybe Hypothesis, and then empirical strategy/preview findings.

Keywords: key1, key2, key3

JEL Codes: key1, key2, key3

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1 Introduction

Ownership changes are increasingly common in healthcare markets and have become a growing concern for policymakers. In U.S. Nursing homes, Skilled nursing facilities (SNFs) can change ownership through acquisitions, sales, and corporate reorganizations, and these changes are frequently talked about because of their possible implications to the quality of care (SOURCES). Between 2016 and 2021 more than 3,200 SNFs were sold in the United States (Prusynski et al., 2023), and this number will continue to increase well into the future (SOURCE). Nursing homes serve medically vulnerable residents who depend on staff for both oversight and daily living assistance, even small changes in staffing could translate into significant differences in care. We study nurse staffing as a central input to the nursing home care production, through which ownership changes may affect care delivery. Specifically we ask, how do ownership changes affect nurse staffing levels in U.S. nursing homes, and do effects differ in chains vs non-chains and before versus during the COVID-19 pandemic?

Nursing homes provide a particularly important setting in which to study the non-price consequences of ownership changes. Shortcoming in nursing home quality has been a long standing concern, and the policy debate has repeatedly emphasized incentives and constraints that shape providers' quality choices (Hackmann, 2019). Nursing home markets also exhibit severe frictions: residents and families place substantial weight on proximity, making facility decisions central to the care experienced by local populations (Gupta et al., 2021). These frictions imply that ownership changes can affect resident welfare through internal production decisions even when the facility remains open and continues to serve the same people facing the same regulations. Staffing is an especially natural outcome to study because it is both highly important for quality and one of the few outcomes that can adjust relatively quickly in response to changes in ownership.

Nursing labor is the core input in nursing home care production, and staffing is one of the most consequential operation choices facilities make.We distinguish among registered nurses (RNs), licensed practical nurses (LPNs)and certified nursing assistants (CNAs). RNs include registered nurses and directors of nursing and play central supervisory and clinical roles, including assessing residents' conditions, developing and coordinating care plans, overseeing medication-related processes, and supervising other staff. LPNs provide routine clinical services and medication management tasks such as monitoring vital signs, while CNAs deliver the majority of hands-on daily care, assisting with bathing, dressing, toileting, feeding, and mobility (Lin, 2014). Because these staff types differ in training, scope of practice, and cost, ownership changes may affect not only total staffing hours but also the skill mix of care delivered. For example, an owner seeking quick cost containment may reduce total hours or substitute away from higher-cost licensed nurses, while an owner focused on compliance or clinical quality may increase RN oversight or adjust supervision structures.

A large body of evidence indicates that staffing is directly linked to nursing home quality. Early work shows that fewer RN hours and fewer aide hours are associated with more deficiencies, including deficiencies related to quality of care and quality of life (Harrington et al., 2000). Importantly, however, staffing is an endogenous choice made jointly with other inputs and under a set of constraints such as regulation and budgets that complicate causal interpretation of simple staffing and quality correlations.

Paragraph on Evidence for why staffing is so important to quality

Our focus on staffing responses to ownership changes is also motivated by literature which shows that financial incentives and ownership structure can have an influence care in nursing homes. Private Equity (PE) ownership of nursing homes has been a rising policy concern, and PE ownership incentives can affect patient outcomes and operational decisions. Gupta et al. (2021) find large increases in short-term mortality associated with PE owned nursing homes and show that staffing composition changes following PE acquisition. They find declines in LPNs and CNAs alongside increases in RN hours, highlighting staffing as a mechanism linking ownership incentives to resident outcomes. While private equity represents one specific ownership form, these findings prove a more general point that is central to this paper: changes in ownership can shift objectives and constraints in ways that reshape internal production decisions, including both staffing levels and skill mix.

Research around how staffing responds to ownership changes is mixed and can be ambiguous. Ownership changes can alter managerial objectives, how owners trade off margins against quality, compliance, and reputation, and can also change the financial environment facing a facility. On one hand, new owners may face stronger incentives to reduce operating costs due to tighter budget targets or financing obligations, making labor cost reductions attractive in a sector where staffing represents a large share of operating expenses. Ownership changes may also introduce short-run disruptions such as turnover in administrators, changes in scheduling and staffing policies, contract renegotiations, and changes in recruitment and retention practices. These channels point toward reductions in staffing or substitution toward lower-cost inputs. On the other hand, ownership changes could increase staffing if new owners place greater weight on quality and compliance, seek to improve performance, or attempt to protect reputation and occupancy in a heavily monitored sector. Public reporting and quality oversight can heighten the incentive to invest in quality and staffing, particularly when quality is noticeable and easily measured (Park and Werner, 2011). In this framework, ownership changes can plausibly generate either quality reducing cost containment or quality improving investment, and the direction may differ by staff type.

This cost–quality tradeoff is highlighted by the fact that nursing home revenues are mostly fixed in the short run. Nursing homes are heavily reliant on public payers, and limited pricing flexibility can constrain the ability to finance staffing investments. Gupta et al. (2021) emphasize this reliance on subsidy and market frictions, and Bowblis and Brunt (2025) give direct evidence that financial resources are tightly linked to staffing: facilities with high Medicaid payer mix have lower staffing levels and staffing expenditures, consistent with the idea that inadequate reimbursement and thin margins limit the ability to hire and retain nursing staff.

More broadly, ownership transitions can also affect the cost structure of facilities through channels that do not translate into patient care investment. Holmes (1996) shows that ownership changes are associated with higher capital related costs, plant costs, interest costs, and taxes, without corresponding increases in patient-care expenditures, implying that facility sales may shrink operating margins and increase pressure to cut variable costs such as staffing. Regulation can further shape how facilities adjust: in response to minimum quality standards, facilities may increase staffing but offset the cost through reductions in other inputs, and they may substitute across nurse types in ways that differ across settings (Bowblis and Ghattas, 2017). This institutional setting implies that staffing could rise, fall, or shift in composition following ownership change, making the net effect an empirical question and motivating a design that can detect these ambiguous adjustments.

This paper contributes to several related literatures. First, it adds to the growing body of work on ownership changes and provider performance by focusing on internal production decisions, staffing, rather than solely on market structure or prices. Second, it contributes to the nursing home literature by studying how a major organizational event, an ownership change, effects the central input into care, building on evidence that staffing improves quality (Lin, 2014) and that staffing shortfalls are linked to poor regulatory performance (Harrington et al., 2000). Third, it complements work on financial incentives and quality in nursing homes by explicitly connecting ownership transitions to staffing adjustments in an environment where financial performance can shape quality investment (Park and Werner, 2011) and where Medicaid reliance constrains staffing resources (Bowblis and Brunt, 2025). By examining heterogeneity by chain affiliation and by period relative to COVID-19, the paper sheds light on how organizational structure and shifting labor market conditions mediate staffing responses to ownership change.

To answer the research question, we use Centers for Medicare and Medicaid Services (CMS) date to construct a panel of nursing homes across the 48 contiguous U.S. states from January 2017 through June 2024. We measure staffing outcomes using hours per patient day (HPPD) for RNs, LPNs, CNAs, and total nursing hours. We estimate treatment effects using a staggered adoption difference-in-differences event study design. The results show persistent declines in nursing staffing following ownership changes, concentrated among RNs, CNAs, and total nursing hours, with heterogeneity by chain affiliation and the COVID-19 pandemic.

2 Data and Methods

2.1 Data Sources

We utilize three main sources of data: Nursing Home Compare (NHC) Archive data, Healthcare Cost Report Information System (HCRIS), and Payroll Based Journal (PBJ) data. The NHC Archive contains monthly facility level information on ownership, chain affiliation, certification status, and quality measures. The second is HCRIS, also known as the Medicare Cost Reports, which contains annual cost reports submitted by every Medicare certified nursing home in the United States. The cost reports contain information about costs, revenue, financial data for each reporting facility, and dates of ownership changes. Lastly, we use the PBJ data to obtain staffing information. The PBJ data contains the census and the number of hours worked by various types of NH staff on a daily basis.

2.2 Sample selection

Our sample is constructed at the facility-year-month level. We create our sample using the following criteria. First, we limit our sample to the 48 contiguous U.S. states from the period January 2017 to June 2024. Secondly, the NHC Archive data is matched to the HCRIS data on a unique nursing home provider number. Because HCRIS only contains information on freestanding nursing homes, our sample excludes hospital-based nursing homes. Hospital based nursing homes differ from freestanding facilities in their structure and financing. Because they are within hospitals, staffing policies and resource allocation may be jointly determined with hospital operations and respond to hospital shocks and regulations rather than nursing home specific incentives. We therefore exclude hospital based facilities to maintain a more homogeneous set of providers and to ensure ownership transitions reflect changes in nursing home management rather than broader hospital system changes. Hospital based nursing homes make up about 12 percent of all nursing homes in the United States.

2.3 Identifying Ownership Changes

No single data source provides a comprehensive record of ownership transitions for U.S. nursing homes, so we combine information from HCRIS and the NHC Archive in a two-step identification and verification process to identify ownership changes.

The first step was to use the HCRIS data which contains an indicator for an ownership change, and corresponding date, in each cost report after an ownership change. We use this as our flag for ownership changes during the study period. However, these indicators may capture administrative or legal changes that do not reflect meaningful transfers of control. Thus, to verify these changes, our second step was to validate each potential ownership change in the HRCIS data with the NHC Archive data. The NHC Archive data reports ownership shares for each owner associated with a facility on a monthly basis. One issue with the NHC Archive data is that it reports the direct owners and indirect owners of each NH. For example, a limited liability corporation may directly own the NH, but that limited liability corporation is owned by different individuals. Therefore, we identified ownership changes to occur among direct and indirect owners using the following criteria:

Using this information, we define a change in ownership when a majority of ownership shares turn over between month $t-1$ and month t. Intuitively, this captures transactions in which control of the facility changes hands, including cases where a new operator acquires most or all shares. Secondly, when ownership percentages are partially or completely missing, we approximate shares by normalizing any reported percentages and, where necessary, assigning equal weights across remaining owners. To reduce false positives from purely legal restructuring, we do not flag a change in ownership if at least 80 percent of the ownership stake remains within the same surname family (e.g., a change from "Smith Holdings LLC" to "Smith Family Partners LP").

Using these verified ownership changes, the analytic sample included Nursing Homes that either (i) have no ownership change in either source or (ii) have exactly one ownership change in each source with the ownership change date within six months of one another. For these Nursing Homes, we define the change in ownership month as the first month in which the NHC Archive data indicate a change. All dates are converted to calendar months, so changes occurring at the beginning or end of a month are treated identically.

2.4 Nursing Staff Levels

Our main dependent variables are measures of nursing staff levels. The nursing staff types we use in our study are registered nurses (RNs), licensed practical nurses (LPNs), and certified nursing assistants (CNAs). RNs are the most highly trained nursing staff and are responsible for assessment, care planning, medication administration, and supervision of other nursing personnel. LPNs provide basic medical care under the supervision of RNs and physicians, including monitoring vital signs and administering certain treatments. CNAs provide the majority of direct daily care to residents, assisting with activities of daily living. Because these staff types differ in training, scope of practice, and compensation, changes in staffing composition may reflect different organizational responses to ownership changes. We therefore examine each staff category separately, as well as total nursing hours per patient day.

Nursing staff levels were calculated from PBJ data, which reports the number of hours worked and the census for each type of nursing staff on a daily basis. We used this information to calculate nursing staff levels in hours per patient day (HPPD) as follows: for each facility i, calendar month t, and staff type, we first aggregate daily hours to the monthly level and then compute

$$\text{HPPD}_{i,t} = \frac{H_{i,t}}{R_{i,t}}, \tag{1}$$

where $H_{i,t}$ is the total hours worked by the staff type during month t and $R_{i,t}$ is the total resident days in that facility-month. Total HPPD is defined as the sum of RN, LPN, and CNA HPPD.

We construct two additional measures of data quality. First, we define a monthly coverage ratio

$$C_{i,t} = \frac{\text{DaysReported}_{i,t}}{\text{DaysInMonth}_t},$$

where $\text{DaysReported}_{i,t}$ is the number of distinct days in month t with complete PBJ records for facility i. This measure allows us to identify incomplete data in our sample and describe our data completeness. We use this measure for sample construction and descriptive purposes, excluding facilities and observations with low reporting coverage.

Second, we track gaps in the reporting for each facility. For each observation we compute *gap*, defined as the number of calendar months since the previous PBJ observation for that facility. This captures discontinuities in reporting that may arise from administrative faults or temporary periods of not reporting PBJ system, for example during the COVID-19 period. Our baseline specification places no restriction on reporting gaps, however; we use the gap measure in robustness analyses to assess whether our results are sensitive to excluding observations following reporting gaps.

Finally, we exclude a small number of observations with implausible staffing levels: months in which RN and LPN HPPD are both zero, total HPPD is below 1.5 or above 12, or CNA HPPD exceeds 5.25. These observations are outliers that likely reflect reporting errors rather than genuine variation in staffing. These exclusion criteria align with the criteria outlined in the CMS NHC Technical Users Guide.

2.5 Other Controls

To account for other factors that are correlated with nursing staff levels, we included a number of control variables that are consistent with other studies (XXX). These control variables were measured at the facility-level and came from NHC and HCRIS. The following variables are used as controls in all specifications: Ownership type (government, non-profit, and for-profit), chain affiliation, total number of beds, occupancy rate, payer mix (percent of revenue coming from Medicare, and percent of revenue coming from Medicaid), and acuity. Our measure of acuity is case-mix provided by CMS in the NHC archive data. During our study period CMS altered the methodology used to construct case-mix indices. As a result, raw case-mix values are not directly comparable over time. To address this issue, we measure acuity using relative rankings rather than levels. Specifically, we construct case-mix quartiles within each state and calendar month.

2.6 Analytic Approach

If nursing homes changed at the same time we would use classic diff in diff, but since they don't, we dont. In our empirical setting, nursing homes change ownership at different points in time. Recent literature shows that conventional two-way fixed effects (TWFE) estimators can produce biased estimates in staggered adoption settings when treatment effects are heterogeneous over time (Goodman-Bacon, 2021). In particular, TWFE estimates can place negative weights on certain comparisons and implicitly rely on already treated units as controls. To circumvent this, we employ a staggered adoption event study regression design. This approach allows us to estimate treatment effects before and after ownership changes while assessing pre-trends. Specifically we estimate the following event study regression:

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_\tau \, \mathbf{1}\{\text{event\_time}_{it} = \tau\} \cdot \text{Treat}_i + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}. \tag{2}$$

where Y_{it} denotes nursing staff levels in HPPD for facility i in in month t. The variable event_time_{it} measures the number of months relative to the month of ownership change. Thus, the coefficients β_τ capture the evolution of the nursing staff levels before and after ownership transitions. The vector X_{it} includes time-varying facility characteristics, while α_i and λ_t denote facility and month fixed effects, respectively. Identification comes from within-facility changes in staffing relative to the timing of ownership transitions, using facilities that have not yet experienced a change as controls.

In addition to estimating using levels, we also estimate specifications where the dependent variable is expressed in logarithms, $\log(Y_{it})$. Log specifications allow coefficients to be interpreted as approximate percentage changes in staffing and reduce sensitivity to differences in baseline staffing levels across facilities. For outcomes with zero staffing in a given month, log specifications are omitted.

2.6.1 Parallel Trends

The identifying assumption underlying our event-study design is that, absent an ownership change, staffing outcomes in treated facilities would have followed the same trends as those in untreated facilities. We assess the plausibility of this assumption by examining estimated coefficients on the pre-treatment event-time indicators and by conducting joint tests of pre-trends. Specifically, we test whether the coefficients on all pre-treatment leads of the ownership change are jointly equal to zero using Wald tests.

Table 7 reports p-values for individual pre-treatment event-time coefficients across outcomes, separately for specifications that include all observations (Panel A) and for specifications that exclude the anticipatory period immediately preceding ownership changes (Panel B). Under the parallel trends assumption, we would expect pre-treatment coefficients to be statistically indistinguishable from zero, implying relatively large p-values and no systematic pattern of significance across pre-treatment periods.

Table 1: Joint Wald Tests of Pre-trends (Event Study)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Levels (HPPD)				
2 Year Full Pre-Window	107.49 (23)	41.63 (23)	46.13 (23)	51.22 (23)
	[0.0000]	[0.0100]	[0.0029]	[0.0006]
2 Year Window with Donut	34.83 (20)	36.57 (20)	17.83 (20)	24.92 (20)
	[0.0210]	[0.0132]	[0.5984]	[0.2046]
1 Year Window with Donut	7.55 (8)	10.49 (8)	3.89 (8)	6.08 (8)
	[0.4788]	[0.2322]	[0.8666]	[0.6387]
Panel B: Logs (HPPD)				
2 Year Full Pre-Window	78.21 (23)	45.72 (23)	34.54 (23)	49.22 (23)
	[0.0000]	[0.0032]	[0.0578]	[0.0012]
2 Year Window with Donut	17.14 (20)	37.42 (20)	13.28 (20)	22.26 (20)
	[0.6436]	[0.0104]	[0.8651]	[0.3268]
1 Year Window with Donut	3.33 (8)	13.48 (8)	7.50 (8)	6.33 (8)
	[0.9122]	[0.0965]	[0.4837]	[0.6102]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
Tested windows and reference periods: 2 Year Full Pre-Window tests $\tau = -24$ to $\tau = -2$ with reference $\tau = -1$; 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Window with Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).
Sample sizes (rows): 2 Year Full Pre-Window ($N = 632,231$); 2 Year Window with Donut ($N = 628,107$); 1 Year Window with Donut ($N = 628,107$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

Table 1 reports the results of our Wald test. In the "2 Year Full Pre-Window" specification, which includes the full set of pre-treatment months ($\tau = -24$ to $\tau = -2$) and uses $\tau = -1$ as the reference period, the joint Wald tests reject the null of no pre-trends for several outcomes, particularly for RN and total staffing in both levels and logs. This indicates that staffing shows change effects in the periods leading up to the recorded ownership transition when the months immediately preceding the change are included in the pre-window, consistent with anticipatory adjustments around the transition date.

We therefore also report "donut" specifications that exclude the three months immediately preceding ownership changes ($\tau \in \{-3, -2, -1\}$) and use $\tau = -4$ as the reference period. Relative to the full pre-window tests, the donut specifications yield substantially weaker evidence of pre-trend violations for several outcomes, with joint tests that are no longer statistically distinguishable from zero in the shorter one-year donut window. Overall, these results suggest that the months immediately before the recorded change month are likely contaminated by anticipation, and that excluding $\tau \in \{-3, -2, -1\}$ improves the plausibility of the parallel trends assumption.

Accordingly, our preferred specifications estimate equation (2) excluding observations in $\tau \in \{-3, -2, -1\}$ and interpret dynamic treatment effects relative to $\tau = -4$. Standard errors are clustered at the facility level to account for serial correlation within nursing homes.

While our main specification relies on the event-study design, we also present TWFE estimates to summarize the average post ownership change effect. We estimate the following regression:

$$Y_{it} = \beta_{\text{post}}\, Post_{it} + X_{it}'\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it}, \tag{3}$$

where Y_{it} denotes nurse staffing outcomes for staff types RN, LPN, CNA, or total HPPD for facility i in month t. The indicator $Post_{it}$ equals one for all months following an ownership change and zero otherwise. Facility fixed effects α_i absorb time invariant differences across nursing homes, while month fixed effects λ_t control for common shocks to staffing over time.

The coefficient β_{post} captures the average change in staffing following an ownership transition, relative to the pre-change period. All specifications include the same time varying covariates as in the event-study models, and standard errors are clustered at the facility level.

2.6.2 Subgroup Analysis

We further examine heterogeneity in the effects of ownership change with two further analyses: the COVID-19 pandemic and baseline chain affiliation.

First, we estimate separate models for the pre-pandemic period (January 2017 through December 2019) and the pandemic period (April 2020 through June 2024). Labor market conditions for nursing staff changed substantially during the COVID-19 pandemic, including heightened turnover, changes in wages, and staffing shortages. Estimating models separately by period allows us to assess whether ownership changes had differential effects on staffing before and during this structural shift in the nursing labor market.

Second, we examine heterogeneity by baseline chain status. We classify facilities based on whether they were affiliated with a chain at the beginning of the study period (January 2017), prior to any ownership transitions in our sample. Chain affiliated facilities differ systematically from independent facilities in ways that are directly relevant for staffing decisions. Prior work shows that chains operate internal labor markets, share managerial expertise across facilities, and are better positioned to reallocate staff in response to shocks, potentially lessening the effects of ownership changes on staffing outcomes. In contrast, independent facilities may face tighter local labor constraints and fewer organizational buffers, making staffing levels more sensitive to changes in ownership and management.

By looking at baseline chain status, we compare facilities with similar organizational structures prior to treatment, rather than confounding the effects of ownership change with changes in chain affiliation occurring at the same time. Estimating models separately for chain and non-chain facilities therefore allows us to assess whether ownership transitions have different effects on staffing across organizational structures, and whether chain affiliation dampens or magnifies staffing adjustments following ownership changes.

In both subgroup analyses, we estimate the same specifications as in equations (2) and (3), maintaining consistent controls, fixed effects, and clustering. These analyses shed light on mechanisms and contextual factors that shape the staffing consequences of ownership changes.

3 Results

3.1 Summary Statistics

Table 2 reports summary statistics for the main analysis sample. The final sample consists of approximately 7,806 unique skilled nursing facilities and 632,231 facility-month observations spanning January 2017 through June 2024. Of these facilities,1589 experience an ownership change during the sample period.

Table 2: Summary Statistics

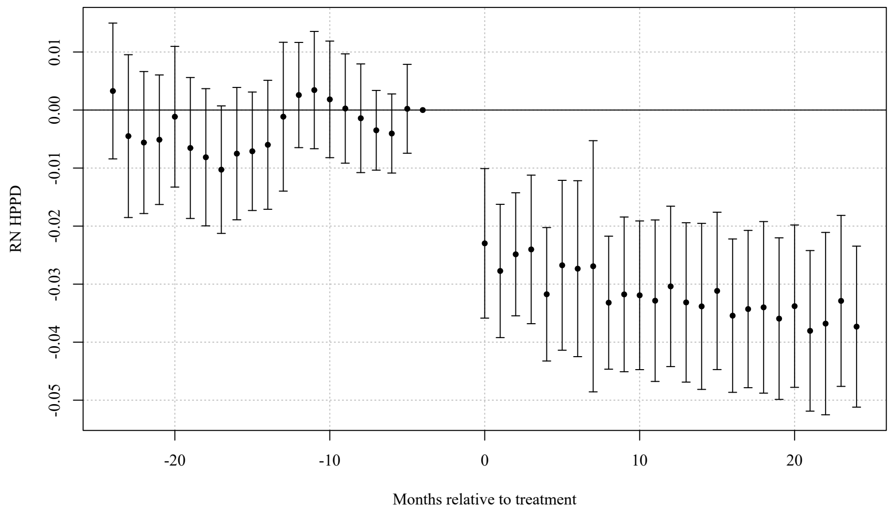
Variable	**Mean**	**SD**
Panel A: Outcome variables		
RN HPPD	0.427	0.309
LPN HPPD	0.796	0.325
CNA HPPD	2.100	0.548
Total HPPD	3.324	0.793
Panel B: Control variables		
Government (dummy)	0.101	0.301
Non-profit (dummy)	0.171	0.377
Chain affiliation (dummy)	0.546	0.498
Beds	108.4	57.9
Occupancy rate (%)	76.4	15.5
% Medicare	13.0	11.1
% Medicaid	60.5	20.2
Acuity quartile 2 (state-month)	0.231	0.422
Acuity quartile 3 (state-month)	0.230	0.421
Acuity quartile 4 (state-month)	0.233	0.423

Notes: The unit of observation is facility–month. HPPD denotes hours per patient day. RN, LPN, and CNA denote registered nurses, licensed practical nurses, and certified nursing assistants, respectively. Total HPPD is the sum of RN, LPN, and CNA HPPD. Occupancy rate is the ratio of residents to certified beds (percent). % Medicare and % Medicaid are the shares of residents covered by Medicare and Medicaid, respectively. Government and Non-profit are ownership-type indicator variables (for-profit omitted category). Chain affiliation is an indicator for membership in a multi-facility chain. Acuity quartiles are state–month quartiles of resident acuity (case-mix) with quartile 1 omitted. Rows = 632,231; Facilities = 7,806; Treated facilities = 1,589; Period = 2017/01–2024/06; Average months per facility = 81.0.

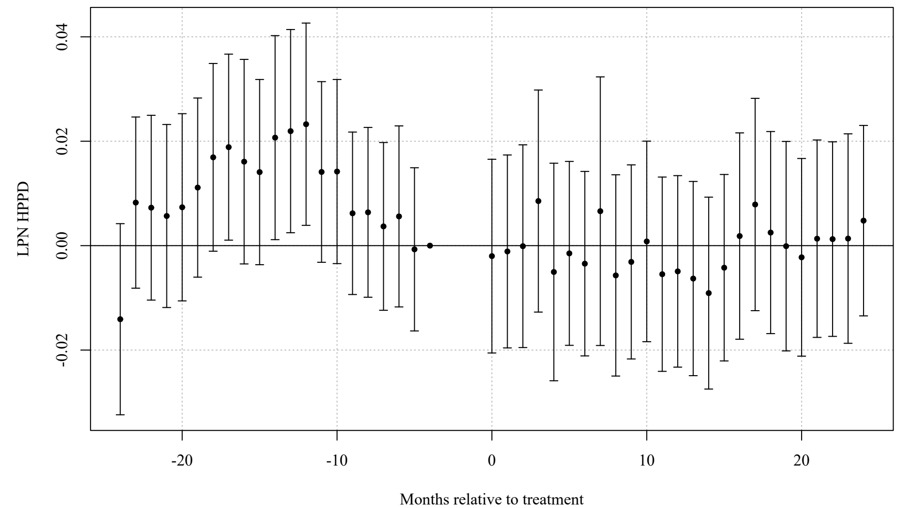
3.2 Main Results

Given our main analysis suggests that parallel trends assumption likely holds, Figure 1 presents event study estimates of the effect of ownership changes on nursing staffing outcomes (i.e., Equation 2). Following an ownership transition, staffing declines persist for many periods, with the largest and most precise effects occurring for registered nurses (RN), certified nursing assistants (CNA), and total nursing hours per patient day.

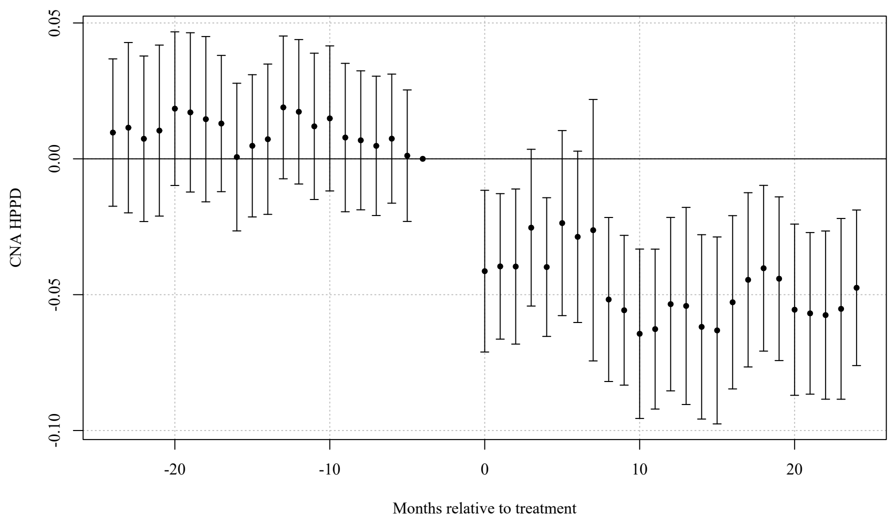
RN staffing declines immediately following ownership changes and remains far below pre-transition levels throughout the post-treatment period. CNA staffing exhibits a similar but slightly smaller decline, while total nursing hours mirror the combined reductions across staff types. In contrast, LPN staffing shows limited and statistically insignificant changes over time. These dynamic patterns indicate that ownership transitions lead to sustained reductions in staffing rather than short run adjustments.



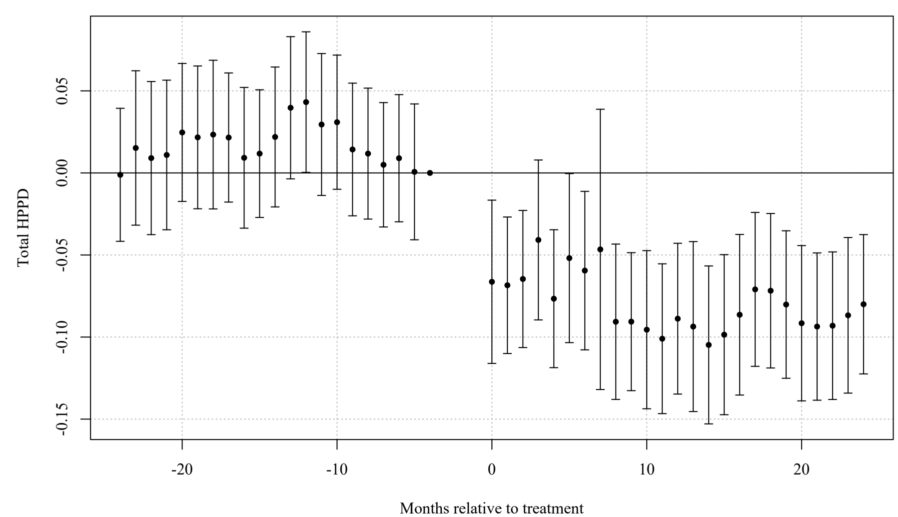
(a) Registered Nurses (RN)



(b) Licensed Practical Nurses (LPN)



(c) Certified Nursing Assistants (CNA)



(d) Total Nursing Hours

Figure 1: Dynamic Effects of Ownership Change on Nursing Staffing. Event study estimates of the effect of a change in ownership on nursing staff levels in hours per patient day (HPPD). Points show coefficient estimates relative to the month immediately preceding the ownership change; vertical bars denote 95% confidence intervals. The three months prior to the ownership change ($\tau = -3, -2, -1$) are excluded. All specifications include control variables and facility and month fixed effects.

To summarize these dynamic effects with a single average treatment estimate, Table 3 reports two-way fixed effects estimates of the ownership change effect (i.e., Equation 3). Panel A reports results in levels, while Panel B reports logged results. Consistent with the event study results, ownership changes lead to meaningful reductions in staffing levels. RN staffing declines by approximately 0.03 hours per patient day (roughly 9 percent), CNA staffing falls by about 0.05 hours (approximately 3 percent), and total staffing decreases by 0.08 hours per patient day (roughly 3 percent). LPN staffing shows no statistically significant change. Estimates are nearly identical whether the observations for the three month period before the ownership change occurred is included or excluded from the model.

Table 3: Two-Way Fixed Effects Estimates of *post* on Staffing Outcomes (Baseline, Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
HPPD	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
	(0.005)	(0.006)	(0.009)	(0.013)
Log(HPPD)	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
	(0.016)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. The table reports staffing levels (HPPD) and log staffing levels (Log(HPPD)). Sample: Without anticipation ($N_{\text{HPPD}} = 628,107$; $N_{\text{Log}} = 628,107$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

3.3 Effects Before and During Covid-19 Pandemic

Table 4 examines whether these effects differ before and after the COVID-19 pandemic. The magnitude of the decline is roughly halved in the post pandemic period for RN, CNA, and total HPPD, suggesting that pandemic induced shifts in labor markets and staffing substantially reduced the impact of ownership transitions.

Table 4: TWFE Estimates of *post*: Pre- vs Post-pandemic Periods (Without anticipation)

	Outcomes			
	RN	LPN	CNA	Total
Panel A: Staffing Levels in HPPD				
Pre-Pandemic Period (2017/01 - 2019/12)	-0.029^{***}	-0.016^{*}	-0.063^{***}	-0.108^{***}
	(0.006)	(0.008)	(0.013)	(0.019)
Pandemic Period (2020/04 - 2024/06)	-0.018^{**}	-0.000	-0.041^{***}	-0.058^{***}
	(0.007)	(0.008)	(0.015)	(0.020)
Panel B: Log Staffing Levels in HPPD				
Pre-Pandemic Period (2017/01 - 2019/12)	-0.068^{***}	-0.009	-0.032^{***}	-0.033^{***}
	(0.024)	(0.012)	(0.006)	(0.006)
Pandemic Period (2020/04 - 2024/06)	-0.063^{***}	0.002	-0.020^{**}	-0.019^{***}
	(0.021)	(0.011)	(0.008)	(0.006)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. Panel A reports levels (HPPD); Panel B reports logs (HPPD). Sample sizes: Row 1 ($N_{\text{levels}} = 255,005$; $N_{\text{logs}} = 255,005$), Row 2 ($N_{\text{levels}} = 359,413$; $N_{\text{logs}} = 359,413$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.
Pre-pandemic 2017/01–2019/12; Pandemic 2020/04–2024/06.

3.4 Effects by Chain Status

Table 5 compares effects for facilities that began the study period as part of a chain and those that did not. Staffing reductions are noticeably larger among non chain facilities. CNA HPPD falls by 0.034 hours in chain facilities and by 0.079 hours in non chains, and total HPPD declines by 0.062 versus 0.113 hours, respectively. On a log scale, these differences translate into substantially larger percentage declines for non-chain facilities. LPN staffing again shows minimal movement.

Table 5: TWFE Estimates of *post*: Chain vs Non-chain Facilities (Jan 2017 Baseline, Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPPD				
Chain January 2017	-0.031^{***}	0.003	-0.034^{***}	-0.062^{***}
	(0.007)	(0.008)	(0.011)	(0.015)
Non-chain January 2017	-0.037^{***}	0.003	-0.079^{***}	-0.113^{***}
	(0.009)	(0.011)	(0.019)	(0.025)
Panel B: Log Staffing Levels in HPPD				
Chain January 2017	-0.099^{***}	0.008	-0.019^{***}	-0.020^{***}
	(0.021)	(0.013)	(0.006)	(0.005)
Non-chain January 2017	-0.085^{***}	0.013	-0.037^{***}	-0.032^{***}
	(0.030)	(0.017)	(0.010)	(0.008)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. Panel A reports levels (HPPD); Panel B reports logs (HPPD). Sample sizes: Row 1 ($N_{\text{levels}} = 322,100$; $N_{\text{logs}} = 322,100$), Row 2 ($N_{\text{levels}} = 243,036$; $N_{\text{logs}} = 243,036$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.
Baseline chain classification determined by facility status in January 2017.

Overall, ownership changes lead to sizable and persistent reductions in nursing staffing, with effects concentrated among RN, CNA, and total hours per patient day. These reductions are larger prior to the COVID-19 pandemic and among facilities that were not part of a chain at baseline. Together, the event study and TWFE results indicate that ownership transitions systematically worsen staffing conditions in nursing homes rather than inducing temporary adjustments.

3.5 Robustness

This section presents a series of robustness checks designed to assess the validity of our identifying assumptions and the sensitivity of our results to alternative model choices. Across all exercises, the estimated effects of ownership change on staffing remain stable in sign, magnitude, and statistical significance. Most of these will have tables/plots that will go in the appendix.

3.5.1 Assumptions

Our main event-study specification is estimated using two-way fixed effects (TWFE) with staggered treatment timing. In this setting, TWFE can implicitly use already-treated facilities as controls for cohorts treated later, which may distort event-study estimates when treatment effects are dynamic or heterogeneous across cohorts. To assess sensitivity to this issue, we re-estimate the event study using a stacked difference-in-differences design.

In the stacked approach, we construct cohort-specific samples for each treatment month g. For a given cohort, treated facilities are those with an ownership change in month g, and the control group consists only of facilities that are never treated or not-yet-treated relative to g (i.e., with treatment dates after g). We restrict each cohort-specific sample to a symmetric event window around g and then stack these cohort datasets into a single analysis file. Estimation includes the same covariates as the baseline specification and fixed effects that absorb facility-level time-invariant heterogeneity and common calendar shocks; standard errors are clustered at the facility level.

Appendix Figures 2 through 5 show that the stacked event-study estimates closely track the baseline TWFE event-study results: the post-transition dynamics and the relative magnitudes across staffing types remain qualitatively unchanged. In addition, Appendix Table **??** reports joint Wald tests of pre-treatment coefficients in the stacked specification (under the donut design used in the baseline), which are consistent with flat pre-trends in the remaining pre-period. Together, these results suggest that the main findings are not driven by treated-as-control comparisons that can arise in TWFE under staggered adoption.

3.5.2 Event Window and Anticipation

As we have previously discussed, because of changes to staffing in anticipation of a formal ownership change, the assumptions of parallel trends is violated. As a result of this, in our baseline specifications we "donut" equation 2 and equation 3 by removing $\tau \in \{-3, -2, -1\}$. To test the validity of removing this set of pre-treatment coefficients, we estimate equations 2 and 3 with different "donut" sets and show that the estimated post-treatment effects remain largely unchanged. Row 6 of Table 6 reports estimates of equation 3 with a wider anticipatory period excluded from the sample. The results from row 6 show no differences in values or statistical significance. Similarly, row 7 reports estimates under a smaller anticipatory window that is excluded from the data, and we see no differences in the results.

3.5.3 Gap Size

Because staffing data is occasionally missing or reported with gaps, we examine whether our results are sensitive to the length of time between consecutive observations. Restricting the sample to facilities with smaller reporting gap shows that estimates are similar to those in the full sample, indicating that data gaps do not drive our findings. Row 2 of Table 6 reports our results for equation 3 with the restriction that facilities with a reporting gap of larger than 6 are excluded from our sample. Compared to the baseline (row 1) this result is identical across staff types and statistical significance. The exact same result is true for row 3 which reports estimates for equation 3 where facilities with a reporting gap of greater than 3 are removed. Row 4 and 5 report estimates where gap greater than 1 or equal to 0 respectively, are removed. Similarly, we see no differences in these results with respect to the baseline.

4 Discussions

Our findings indicate that ownership transitions in nursing homes are followed by reductions in nurse staffing, suggesting that labor inputs constitute a primary method of adjustment following changes in ownership. In a sector characterized by extensive requirements and regulatory oversight, staffing decisions represent one of the most flexible ways through which new owners can alter operating costs and organizational practices. The observed declines in RN, CNA, and total nursing HPPD point to a reallocation of resources away from labor following ownership changes. These patterns show the importance of examining internal production decisions when evaluating the consequences of ownership transitions in regulated healthcare markets.

The timing of these staffing adjustments also gives insight into the underlying effects. Event-study estimates in Figure 1 show that staffing levels begin to decline shortly before the true filing transfer date and remain persistently lower in the following periods. This pattern is consistent with ownership changes being anticipated events rather than sudden shocks. During this period before the change, incumbent or incoming owners may begin restructuring operations or implementing cost-containment strategies in advance of the actual transfer date. The persistence of the reductions after ownership changes suggests that adjustments are not just transitory, but rather longer-term changes in organizational priorities and operating practices.

Labor being an adjustment margin in this context is economically intuitive because nursing labor represents one of the largest operating expenses for nursing homes and is subject to fewer regulation than prices or resident admissions. While minimum staffing requirements exist, facilities retain considerable discretion over staffing levels above the threshold, as well as over workforce composition across skill categories. In contrast, rates are largely fixed through Medicare and Medicaid, and changes in bed capacity or service lines are costly and often constrained by regulations and local market conditions. As a result, ownership changes may translate into staffing changes even when other inputs to production remain unchanged.

The heterogeneity of staffing responses across nurse types also shows how ownership changes affect labor inputs. We find larger and more consistent reductions in RN staffing relative to other nurse categories, with declines also evident among CNAs and in total staffing. This suggests that staffing changes are not uniform across skill levels and may reflect differences in wage costs. RN's require higher wages and play supervisory and clinical roles that may be more easily reallocated following ownership changes, especially in facilities that become integrated into larger organizational structures. In contrast, CNA staffing, while it still declines, may be constrained by direct care needs and minimum staffing standards. The differential response across staff types highlights how ownership transitions can alter not only staffing levels but also labor composition, with potential implications for care delivery.

The magnitude of staffing responses also differs between the pre-pandemic and pandemic periods. While ownership changes are associated with statistically significant reductions in staffing in both periods, these effects are consistently smaller during the pandemic. As shown in Table 4, declines in RN, CNA, and total nursing hours per patient day are larger in the pre-pandemic period than during the COVID-19 period, both in levels and in logs. One interpretation is that the pandemic fundamentally altered labor market conditions in long-term care, limiting nursing homes ability to further adjust staffing in response to ownership changes. Widespread staffing shortages of nursing labor during the pandemic may have constrained owners' capacity to implement additional reductions. At the same time, temporary regulation and heightened scrutiny of nursing home staffing during the pandemic may have dampened the extent to which new owners could reduce labor supply.

Staffing responses also vary across chain-affiliated and non-chain facilities, indicating that organizational structure is important in shaping the labor effects of ownership changes. We find that staffing reductions following ownership changes are substantially larger, almost double in CNA and total staffing, among non-chain facilities than among chain-affiliated nursing homes. One potential explanation is that chain-affiliated facilities may benefit from greater centralization and larger labor pools. In contrast, non-chain facilities may face more binding financial constraints, making labor cost reductions a more immediate and pronounced response to changes in ownership. Additionally, independent facilities may have more limited ability to reallocate staff across locations, leading to sharper staffing reduction. These patterns suggest that ownership transitions may be more disruptive for non-chain facilities, with labor inputs bearing a much larger share of adjustment.

We have seen decreases in staffing levels in almost all the staffing categories that we study, but an important question to ask is whether these declines matter? Do these declines in staffing actually translate to real negative affects on patients? In the baseline model, we estimate declines of -0.032 RN HPPD, -0.054 CNA HPPD, and -0.086 total HPPD following ownership changes (with no statistically significant change in LPN staffing). Converting these magnitudes, -0.032 RN HPPD corresponds to about $0.032 \times 60 \approx 1.9$ fewer RN minutes per resident-day, -0.054 CNA HPPD corresponds to about 3.2 fewer CNA minutes per resident-day, and -0.086 total HPPD corresponds to about 5.2 fewer total nursing minutes per resident-day. While a reduction of a few minutes per resident-day may appear tiny, it aggregates to meaningful staff time at the facility level: for every 100 resident-days, a -0.086 total HPPD effect implies about $0.086 \times 100 = 8.6$ fewer total nursing hours per day. Put differently, this is close to removing one full-time shift of nursing labor each day, sustained over time. In a setting where staffing is one of the main inputs to quality, sustained declines of this size are likely to have real implications for patient care. Figure out percentages then make comments about what this actually means.

Our findings contribute to the literature in several ways. First, they complement the industrial organization literature on ownership changes by providing evidence on non-price outcomes in a setting where prices are mostly fixed. While prior work has emphasized price and market-power effects of mergers and acquisitions, our results highlight how ownership transitions can affect internal production decisions even in the absence of price changes. Second, this study adds to the literature on labor market outcomes following ownership changes. Previous evidence on employment and wage effects has been mixed and largely drawn from manufacturing or administrative settings, so by focusing on healthcare labor in nursing homes, we show that ownership changes can have substantial labor effects in a sector where staffing is directly tied to output quality. Third, our results contribute to the healthcare consolidation literature by identifying staffing as an important way through which ownership changes may influence care delivery, complementing existing studies that focus on prices or spending.

Paragraph on Limitations

Closing

5 Conclusion

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Tables

Table 6: TWFE estimates of *post* (without anticipation).

		Outcomes			
	N	**RN**	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPPD					
(1) Baseline	628,107	-0.032^{***} (0.005)	0.000 (0.006)	-0.054^{***} (0.009)	-0.086^{***} (0.013)
(2) Sample excludes *gap* > 6	627,923	-0.032^{***} (0.005)	-0.000 (0.006)	-0.054^{***} (0.009)	-0.086^{***} (0.013)
(3) Sample excludes *gap* > 3	627,626	-0.032^{***} (0.005)	-0.000 (0.006)	-0.054^{***} (0.009)	-0.086^{***} (0.013)
(4) Sample excludes *gap* > 1	623,585	-0.032^{***} (0.005)	0.000 (0.006)	-0.055^{***} (0.009)	-0.086^{***} (0.013)
(5) Sample excludes *gap* > 0	623,585	-0.032^{***} (0.005)	0.000 (0.006)	-0.055^{***} (0.009)	-0.086^{***} (0.013)
(6) Drop $t \in \{-4, -3, -2, -1\}$	626,729	-0.032^{***} (0.005)	0.000 (0.006)	-0.055^{***} (0.009)	-0.086^{***} (0.013)
(7) Drop $t \in \{-2, -1\}$ only	628,107	-0.032^{***} (0.005)	0.000 (0.006)	-0.054^{***} (0.009)	-0.086^{***} (0.013)
Panel B: Log Staffing Levels in HPPD					
(1) Baseline	628,107	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(2) Sample excludes *gap* > 6	627,923	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(3) Sample excludes *gap* > 3	627,626	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(4) Sample excludes *gap* > 1	623,585	-0.092^{***} (0.016)	0.004 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(5) Sample excludes *gap* > 0	623,585	-0.092^{***} (0.016)	0.004 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(6) Drop $t \in \{-4, -3, -2, -1\}$	626,729	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)
(7) Drop $t \in \{-2, -1\}$ only	628,107	-0.092^{***} (0.016)	0.003 (0.009)	-0.028^{***} (0.005)	-0.026^{***} (0.004)

Notes: Each cell reports the *post* coefficient with two-way clustered standard errors at the facility and month levels. Panel A uses staffing levels (HPPD); Panel B uses logs of HPPD.
All specifications include facility and month fixed effects and controls for ownership (government, non-profit, chain), beds, occupancy rate, payer mix (Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 7: Pre-treatment Event-Time Coefficient p-values by τ (Levels Only)

	Outcomes (HPPD)			
	RN	**LPN**	**CNA**	**Total**
Panel A: With Anticipation				
$\tau = -24$	0.0009	0.3074	0.0226	0.0408
$\tau = -23$	0.0299	0.2654	0.0270	0.0133
$\tau = -22$	0.0575	0.2864	0.0504	0.0298
$\tau = -21$	0.0418	0.3507	0.0475	0.0269
$\tau = -20$	0.0063	0.2533	0.0112	0.0046
$\tau = -19$	0.0522	0.1386	0.0058	0.0027
$\tau = -18$	0.1171	0.0333	0.0110	0.0031
$\tau = -17$	0.1677	0.0303	0.0106	0.0028
$\tau = -16$	0.0625	0.0599	0.0898	0.0117
$\tau = -15$	0.0438	0.0761	0.0597	0.0116
$\tau = -14$	0.0333	0.0208	0.0199	0.0015
$\tau = -13$	0.0020	0.0155	0.0031	0.0002
$\tau = -12$	0.0002	0.0107	0.0018	0.0001
$\tau = -11$	0.0001	0.0771	0.0154	0.0017
$\tau = -10$	0.0006	0.0689	0.0105	0.0011
$\tau = -9$	0.0007	0.3447	0.0309	0.0111
$\tau = -8$	0.0015	0.3051	0.0429	0.0162
$\tau = -7$	0.0063	0.4569	0.0643	0.0322
$\tau = -6$	0.0076	0.3574	0.0241	0.0133
$\tau = -5$	0.0007	0.7619	0.0530	0.0281
$\tau = -4$	0.0002	0.6479	0.0578	0.0209
$\tau = -3$	0.0056	0.8746	0.0783	0.0740
$\tau = -2$	0.0969	0.9108	0.4561	0.4669
$\tau = -1$ (Ref.)	*Ref.*	*Ref.*	*Ref.*	*Ref.*
Panel B: Without Anticipation ($t = -3, -2, -1$ dropped)				
$\tau = -24$	0.5788	0.1295	0.4807	0.9552
$\tau = -23$	0.5263	0.3203	0.4694	0.5222
$\tau = -22$	0.3655	0.4165	0.6315	0.7009
$\tau = -21$	0.3649	0.5216	0.5136	0.6343
$\tau = -20$	0.8508	0.4178	0.1977	0.2468
$\tau = -19$	0.2875	0.2010	0.2497	0.3249
$\tau = -18$	0.1749	0.0649	0.3432	0.3081
$\tau = -17$	0.0665	0.0382	0.3063	0.2788
$\tau = -16$	0.1941	0.1063	0.9627	0.6704
$\tau = -15$	0.1700	0.1182	0.7171	0.5494
$\tau = -14$	0.2872	0.0382	0.6055	0.3096
$\tau = -13$	0.8594	0.0277	0.1558	0.0719
$\tau = -12$	0.5721	0.0192	0.1990	0.0483
$\tau = -11$	0.5003	0.1086	0.3802	0.1784
$\tau = -10$	0.7170	0.1133	0.2711	0.1365
$\tau = -9$	0.9555	0.4310	0.5703	0.4842
$\tau = -8$	0.7648	0.4378	0.5979	0.5589
$\tau = -7$	0.3153	0.6498	0.7126	0.7953
$\tau = -6$	0.2416	0.5228	0.5363	0.6464
$\tau = -5$	0.9556	0.9283	0.9247	0.9748
$\tau = -4$ (Ref.)	*Ref.*	*Ref.*	*Ref.*	*Ref.*

Table 8: Joint Wald Tests of Pre-trends (Stacked Event Study) — Levels

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
2 Year Window with Donut	28.46 (20)	35.10 (20)	20.25 (20)	24.08 (20)
	[0.0990]	[0.0196]	[0.4423]	[0.2388]
1 Year Window with Donut	5.26 (8)	24.35 (8)	6.59 (8)	15.99 (8)
	[0.7300]	[0.0020]	[0.5817]	[0.0425]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.

Tested windows and reference periods: 2 Year Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).

Sample sizes (stacked rows): 2 Year Donut ($N = 23,877,687$); 1 Year Donut ($N = 13,121,085$).

All specifications include stacked-unit fixed effects (facility-by-cohort), calendar-month fixed effects, cohort fixed effects, and baseline covariates. Standard errors are clustered at the facility level.

Figures

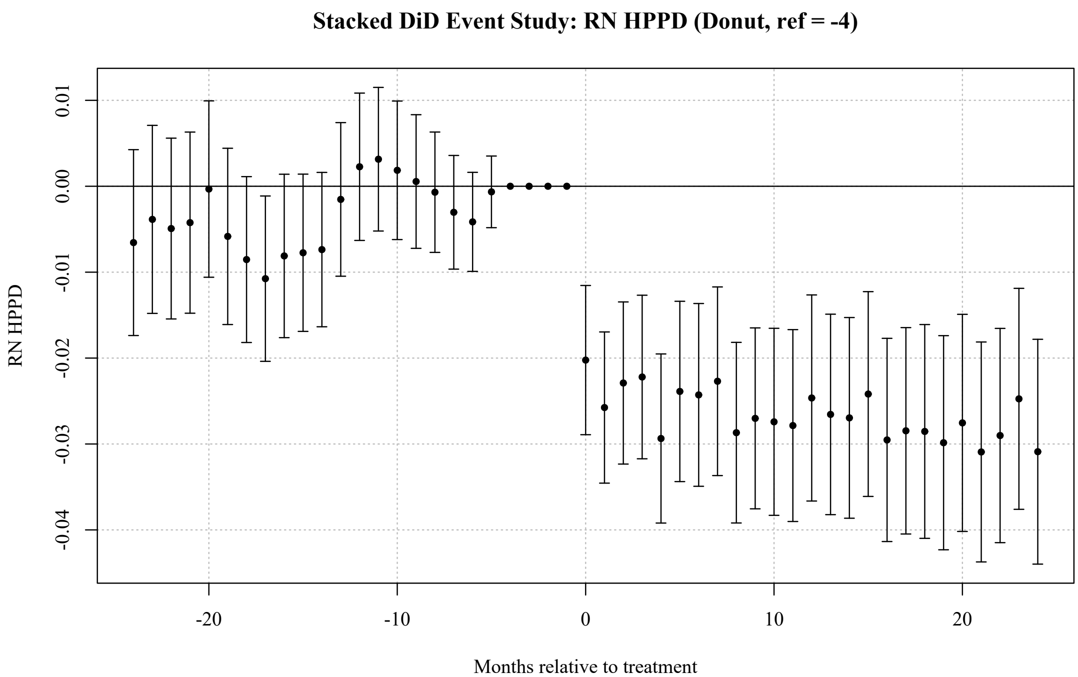

Figure 2: Stacked event-study estimates (RN staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

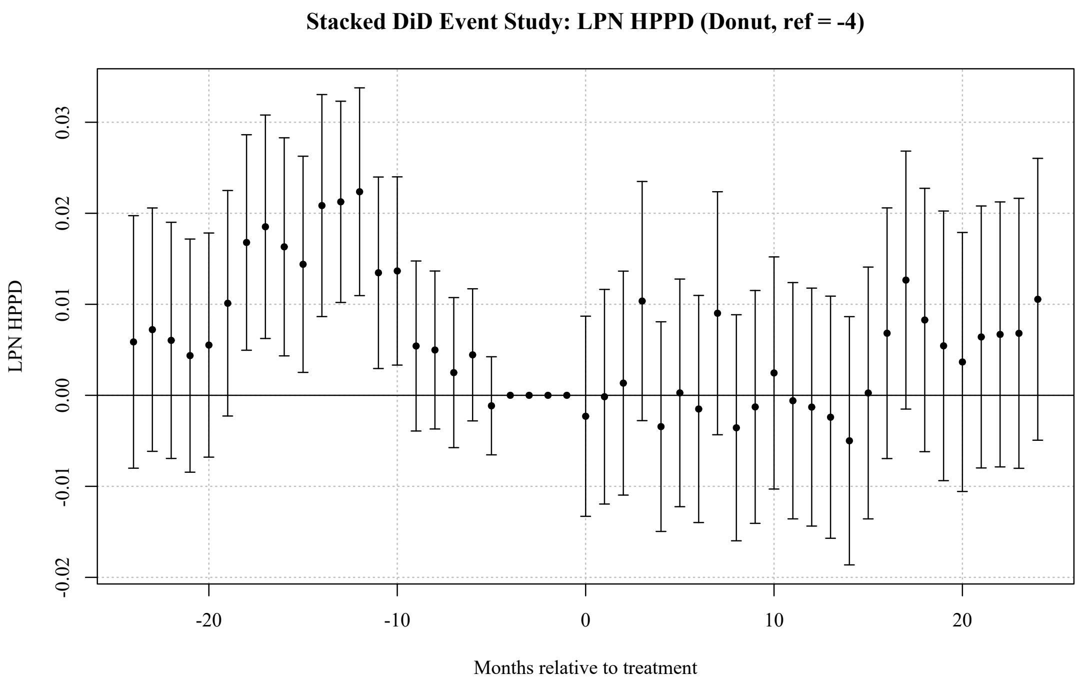

Figure 3: Stacked event-study estimates (LPN staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

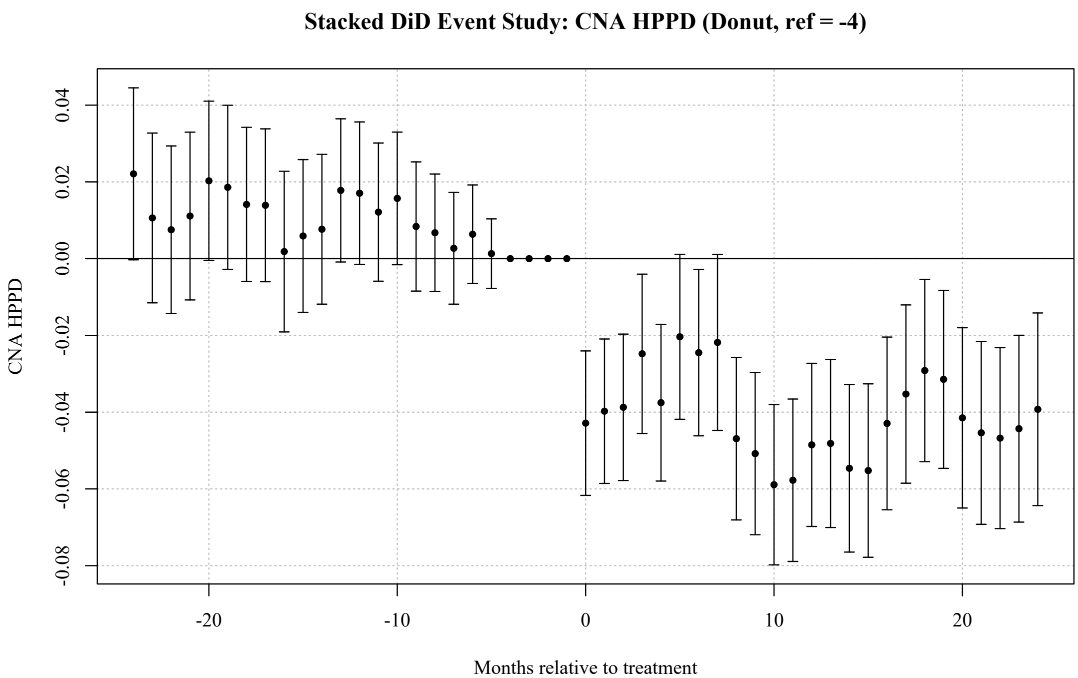

Figure 4: Stacked event-study estimates (CNA staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

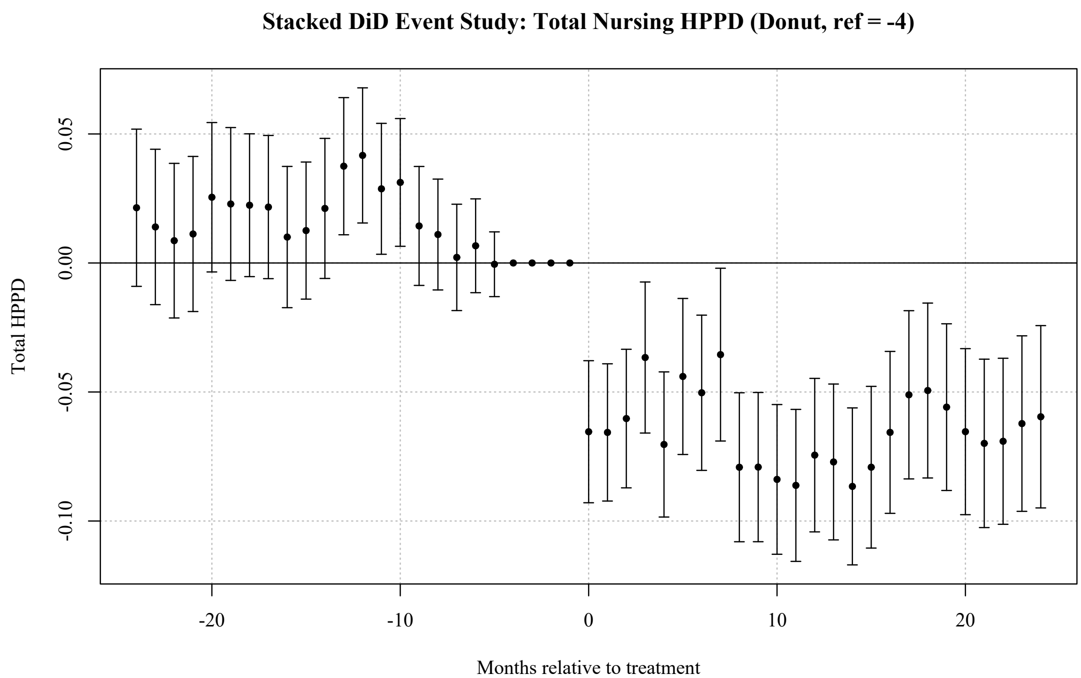

Figure 5: Stacked event-study estimates (Total staffing, levels). The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

Appendix A. Placeholder